

Project 4

Report on

Title: Exploratory Review of Biometric Indicators to Understand Diabetes

Risk

By

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Introduction to Analytics

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1. Introduction

This project delves into a rich medical dataset focused on 768 female patients, all at least 21 years old, from the Pima Indian community. The primary goal is to explore the factors that may contribute to the likelihood of being diagnosed with diabetes. By analyzing the relationships between a range of health indicators and diabetes status, we aim to uncover patterns and potential risk factors that could inform medical screening and intervention strategies.

The dataset encompasses several key biometric and clinical measurements commonly used in diabetes diagnosis and monitoring. These include:

- ♦ **Number of pregnancies** – A demographic factor that may influence metabolic health.
- ♦ **Plasma glucose concentration** – A critical marker of blood sugar levels, often used in diabetes screening.
- ♦ **Diastolic blood pressure** – Represents the pressure in arteries between heartbeats, with high values potentially linked to chronic conditions.
- ♦ **Triceps skinfold thickness** – An estimate of subcutaneous fat, often used to assess body fat distribution.
- ♦ **2-Hour serum insulin** – Measures insulin levels two hours after glucose intake, providing insight into insulin resistance.
- ♦ **Body Mass Index (BMI)** – A standard measure of body fat based on weight and height.
- ♦ **Diabetes pedigree function** – A score that reflects the individual's genetic predisposition to diabetes based on family history.
- ♦ **Age** – As a general health factor, age can influence the risk of developing chronic illnesses, including diabetes.
- ♦ **Diabetes status** – A binary variable indicating whether the individual tested positive or negative for diabetes.

This dataset serves as a valuable resource for understanding the medical and demographic variables associated with diabetes. It not only aids in recognizing significant health patterns but also builds foundational knowledge for applying predictive analytics in healthcare contexts. By interpreting these relationships, we can better support early detection, prevention strategies, and informed healthcare decisions.

2. Data Cleaning and Preparation

Steps executed in R:

- ♦ **Step 1:** Loaded and renamed column headers using `clean_names()` from the `janitor` package.
- ♦ **Step 2:** Replaced biologically invalid zero values (e.g., in glucose, BMI, insulin, etc.) with `NA`.
- ♦ **Step 3:** Converted the diabetes column to a factor for categorical handling.
- ♦ **Step 4:** Removed rows with missing values using `na.omit()`.
- ♦ **Step 5:** Verified structure and summary statistics with `glimpse()` and `summary()`.

3. Feature Engineering

To enrich the analysis and uncover deeper insights from the dataset, additional features were thoughtfully engineered based on existing variables. These derived features help simplify complex numerical data into more interpretable categories, making it easier to identify trends across different segments of the population.

- ♦ **Age Group:** Instead of analyzing age as a continuous variable, individuals were grouped into three distinct life stages:
 - **Under 30** – Younger adults who may still be in early adulthood with potentially lower health risks.
 - **30 to 49** – Middle-aged adults who typically start showing more health-related variations.
 - **50 and above** – Older individuals who are generally more susceptible to chronic conditions like diabetes.
 - This segmentation allows for easier comparison of diabetes prevalence and health metrics across different age brackets.
- ♦ **BMI Category:** Body Mass Index (BMI), which measures body fat based on height and weight, was transformed into four meaningful categories based on widely recognized health standards:
 - **Underweight** – BMI less than 18.5
 - **Normal** – BMI between 18.5 and 24.9
 - **Overweight** – BMI between 25 and 29.9
 - **Obese** – BMI of 30 or higher

- Categorizing BMI this way helps identify whether higher body fat levels correlate with increased diabetes risk.

4. Summary Statistics

This section explores key relationships within the dataset using descriptive statistics and grouped comparisons. By breaking down the data across age groups, diabetes status, and body composition metrics, we gain a clearer understanding of the patterns linked to diabetes risk.

4.1. Average BMI by Age Group and Diabetes Status

age_group	diabetes	avg_bmi
30–49	Healthy	32.72857
30–49	Sick	35.05410
50+	Healthy	30.26667
50+	Sick	34.14762
Under 30	Healthy	31.49684
Under 30	Sick	37.41042

Table 1

The analysis revealed a consistent trend: individuals diagnosed with diabetes had higher average Body Mass Index (BMI) values than those without the condition, regardless of age group.

- ◆ Among patients **under 30**, those with diabetes already showed a noticeably higher BMI than their non-diabetic peers.
- ◆ In the **30–49** and **50+** age groups, the average BMI not only remained higher among diabetics but also increased progressively with age.

This pattern suggests that excess body weight, particularly as people age, is a strong contributing factor to diabetes prevalence.

4.2. Average Glucose by Age Group

age_group	avg_glucose
30-49	129.2016
50+	154.6667
Under 30	115.1639

Table 2

Plasma glucose concentration, a vital indicator for diagnosing diabetes was also examined across age groups.

- The data showed that individuals aged **50 and above** had the **highest average glucose levels**, with those diagnosed with diabetes recording significantly elevated values.
- Even in younger groups, diabetic individuals had higher glucose concentrations, but the gap widened noticeably with age.

This trend underscores the importance of monitoring blood sugar levels, particularly in older adults who may be at greater risk of undiagnosed diabetes or prediabetes.

4.3. BMI Category vs. Diabetes Status

avg_glucose	avg_bmi	avg_age	diabetes_rate
122.6276	33.08622	30.8648	0.3316327

Table 3

When BMI values were categorized into standard weight classes (Underweight, Normal, Overweight, and Obese), a striking association emerged:

- A **majority of diabetic patients** fell into the **Overweight or Obese** categories.
- Very few individuals in the Normal or Underweight categories were diagnosed with diabetes.

This reinforces the well-established link between excess body weight and the development of type 2 diabetes, highlighting the importance of weight management in prevention strategies.

4.4. Mean Insulin and Triceps Skinfold Thickness

bmi_category	Healthy	Sick
Normal	42	2
Obese	152	110
Overweight	67	18
Underweight	1	0

Table 4

While insulin and triceps skinfold thickness data had a considerable number of missing values, average comparisons still revealed subtle insights:

- Diabetic individuals tended to have **slightly higher average insulin levels**, indicating possible insulin resistance.
- **Triceps skinfold thickness**, a proxy for subcutaneous fat, was also somewhat higher among diabetics.

Though the results here are less definitive due to data limitations, these trends are still aligned with expectations and suggest that body fat distribution and insulin activity both play roles in diabetes risk.

4.5. Mean Insulin and Triceps Skinfold Thickness by Diabetes Status

diabetes	mean_insulin	mean_thickness
Healthy	130.8550	27.25191
Sick	206.8462	32.96154

Table 5

This table compares average insulin levels and triceps skinfold thickness between individuals diagnosed with diabetes and those without the condition.

- **Insulin Levels:** People with diabetes showed noticeably higher average insulin levels (206.85 $\mu\text{U/mL}$) compared to non-diabetic individuals (130.86 $\mu\text{U/mL}$). This difference reflects a common characteristic of type 2 diabetes insulin resistance where the body struggles to process glucose effectively, often resulting in elevated insulin production.

- ♦ **Triceps Skinfold Thickness:** Diabetic individuals also had a higher average skinfold thickness (32.96 mm) than their healthy counterparts (27.25 mm). This measurement, which estimates subcutaneous fat, suggests that increased body fat may be linked to a higher risk of developing diabetes.

While both indicators provide valuable insights, it's important to recognize that the dataset had several missing values for insulin and triceps measurements. Therefore, these averages may not fully capture the characteristics of the entire population. However, the overall trends still support established associations between higher body fat, insulin levels, and diabetes risk.

5. Visualizations and Interpretation

5.1. Histogram: BMI by Diabetes Status

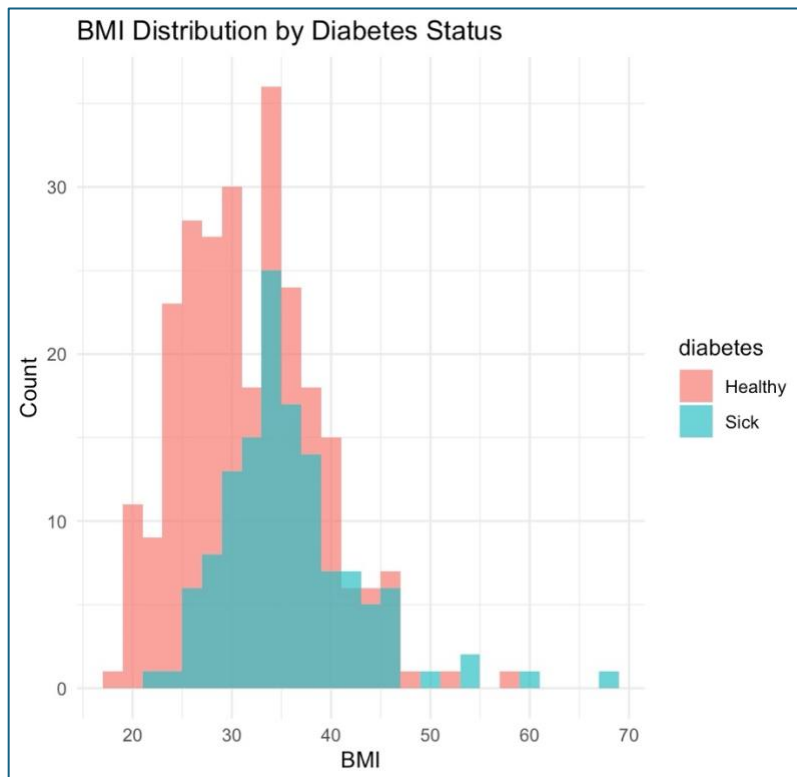


Fig 1

- ♦ The histogram visually compares BMI distributions between diabetic and non-diabetic individuals.
- ♦ A **rightward shift** was observed in the diabetic group, indicating a tendency toward **higher BMI values**.
- ♦ This suggests that individuals diagnosed with diabetes are more likely to fall into the **overweight or obese categories**, reinforcing the strong link between body weight and diabetes risk.

5.2. Scatter Plot – Glucose vs. Age

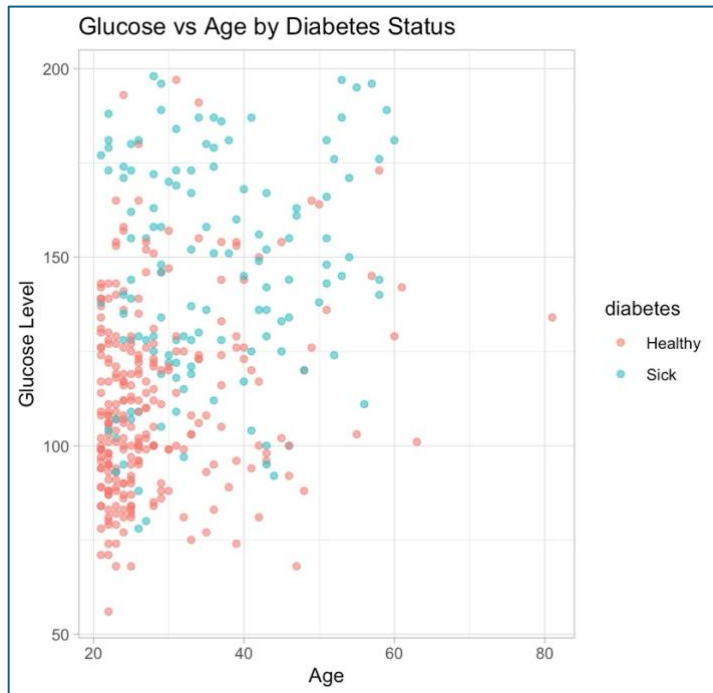


Fig 2

- ♦ This scatter plot displays glucose levels against age for all individuals.
- ♦ Among those with diabetes, a **positive correlation** was noticeable, **glucose levels increased with age**.
- ♦ The trend implies that **older diabetic individuals** tend to have **elevated blood glucose**, underscoring the importance of age as a risk factor and the need for glucose monitoring as people grow older.

5.3. Boxplot – Age by Diabetes Status

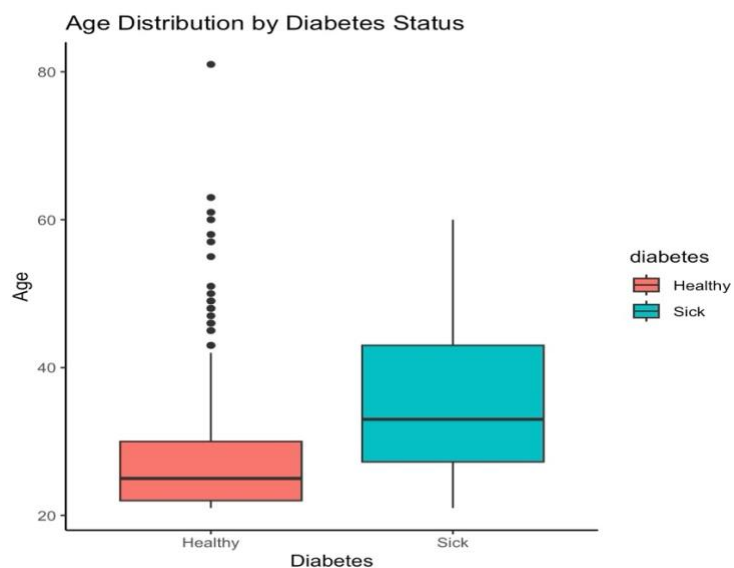


Fig 3

- ♦ The boxplot compares the **age distributions** of diabetic vs. non-diabetic individuals.
- ♦ Diabetic individuals had a **higher median age** and an age range skewed toward older values.
- ♦ This indicates that **diabetes is more prevalent among older individuals**, supporting the idea that age contributes significantly to diabetes susceptibility.

5.4. Bar Plot – Average BMI by Age Group & Diabetes Status

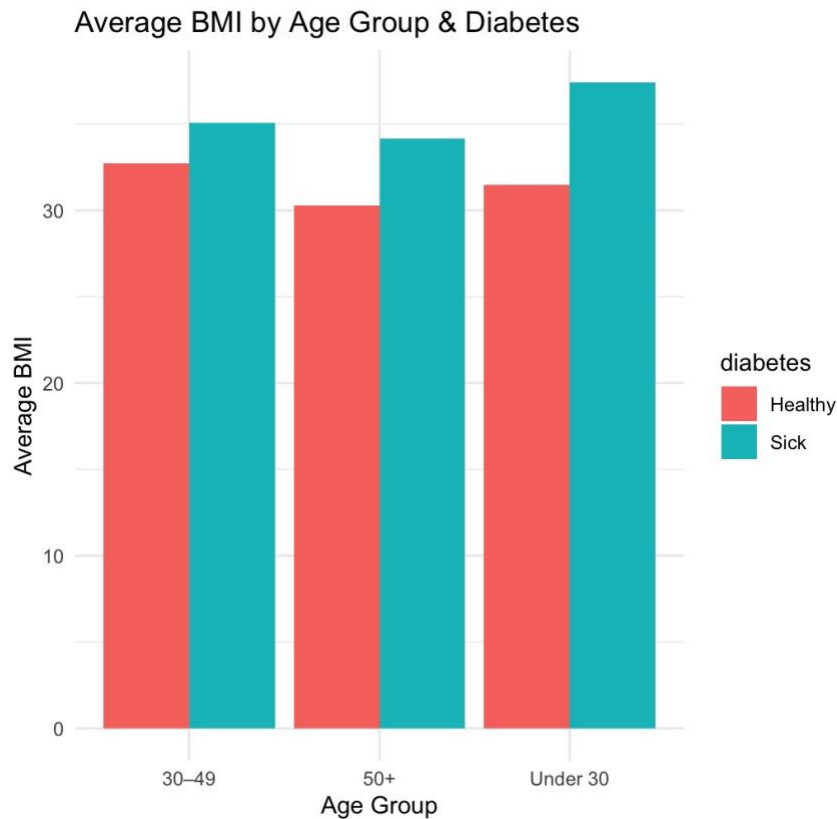


Fig 4

- ♦ This bar chart breaks down **average BMI by age group**, further split by diabetes status.
- ♦ The **50+ diabetic group had the highest average BMI**, showing a combined effect of **age and obesity** on diabetes risk.
- ♦ Even in younger age groups, diabetics maintained a **consistently higher BMI** than non-diabetics, emphasizing that weight is a critical factor across all ages.

6. Key Observations and Insights

The following observations capture the most important patterns identified through both the statistical summaries and visualizations. These insights help highlight the health indicators most associated with diabetes in the dataset:

- ♦ **Higher Glucose and BMI Among Diabetics:** Diabetic individuals consistently exhibited **elevated average glucose levels and BMI**, suggesting that blood sugar regulation and body weight are two of the strongest indicators of diabetes risk.
- ♦ **Age as a Critical Risk Factor:** The data showed that individuals aged **50 and above** had the highest proportion of diabetes cases. This reinforces the notion that **the risk of developing diabetes increases with age**.
- ♦ **Dominance of Overweight and Obese Categories:** The **majority of diabetic individuals** fell into the **Overweight or Obese BMI** categories. This highlights the direct relationship between **excess body fat and diabetes prevalence**.
- ♦ **Glucose Trends with Age:** A **positive trend between age and glucose levels** was observed, particularly among diabetic patients. This suggests that **age may amplify the body's difficulty in managing glucose levels**, increasing diabetes risk over time.
- ♦ **Variations in Insulin and Skinfold Thickness:** Although these metrics had more missing values, diabetic individuals showed **higher average insulin and skinfold thickness**, indicating **underlying physiological differences** in fat storage and insulin regulation.

7. Future Exploration Questions

While this analysis focused on uncovering patterns and associations, it also opens the door to deeper investigation. Some compelling future questions include:

- ♦ **Can we build a predictive model** using variables like **age, glucose levels, and BMI** to estimate an individual's risk of developing diabetes?
- ♦ **What is the predictive value of pregnancy count** in determining diabetes risk among women? Could this demographic feature be more influential than expected?
- ♦ Could a **combined analysis of insulin, BMI, and glucose** lead to improved prediction models, especially for early detection?
- ♦ **How do BMI and glucose interact across age groups?** Is there a tipping point at which the combination of age, weight, and blood sugar creates a significantly higher risk?

These questions could be explored using machine learning models, regression techniques, or multivariate analysis in future studies.

8. Conclusion

This analysis provided a foundational understanding of how certain biometric and health-related variables are linked to diabetes risk within the Pima Indian female population. The patterns observed, **higher glucose levels, increased BMI, and older age among diabetic individuals** are consistent with known risk factors in medical literature.

Variables such as **insulin levels and triceps skinfold thickness**, although limited by missing data, still hinted at meaningful physiological differences between diabetic and non-diabetic individuals. These secondary indicators could become more useful with a more complete dataset.

Overall, this project demonstrates how **basic exploratory data analysis (EDA) using R** can uncover valuable insights in a healthcare context. It shows the potential for **predictive modeling and targeted screening**, particularly in communities at high risk. It also emphasizes the importance of comprehensive, high-quality data to strengthen future research and decision-making in medical analytics.

With continued refinement and modeling, this dataset could serve as the foundation for developing tools that **proactively identify at-risk individuals** and support early intervention strategies, ultimately improving outcomes in diabetes care and prevention.

References

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